

Title

The ChatGPT Effect: Nursing Education and Generative Artificial Intelligence

Authors

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Abstract

This article examines the potential of generative artificial intelligence (AI), such as ChatGPT (Chat Generative Pre-trained Transformer), in nursing education and the associated challenges and recommendations for their use. Generative AI offers potential benefits such as aiding students with assignments, providing realistic patient scenarios for practice, and enabling personalized, interactive learning experiences. However, integrating generative AI in nursing education also presents challenges, including academic integrity issues, the potential for plagiarism and copyright infringements, ethical implications, and the risk of producing misinformation. Clear institutional guidelines, comprehensive student education on

generative AI, and tools to detect AI-generated content are recommended to navigate these challenges. The article concludes by urging nurse educators to harness generative AI's potential responsibly, highlighting the rewards of enhanced learning and increased efficiency. The careful navigation of these challenges and strategic implementation of AI is key to realizing the promise of AI in nursing education.

Editorial

The rapid evolution of artificial intelligence (AI) has birthed groundbreaking tools, including generative language models like OpenAI's ChatGPT (<https://chat.openai.com/chat>). These tools exploit AI's ability to generate fluent and coherent-sounding text mimicking human language—a previously challenging task. In essence, ChatGPT is an AI model trained on a very large body of text extracted from the Internet. As a result, ChatGPT (Chat Generative Pre-trained Transformer) can generate texts that are hardly distinguishable from human-produced texts (Dwivedi et al., 2023; Ray, 2023). In addition, ChatGPT was shown to have information on multiple disciplines, exemplified by its ability to successfully pass various professional exams, including medical and nursing licensure exams (Achiam et al., 2023).

We are seeing the impacts of these AI technologies on facets of everyday life. The field of education has been among those most impacted by the release of ChatGPT, spurring widespread concerns and conversations related to pedagogy and evaluation of learning. From AI tutors to smart classrooms, AI-based technologies are starting to augment teaching and learning experiences in novel ways. Among various disciplines, nursing, specifically nursing education, is beginning to recognize generative AI applications' potential benefits and drawbacks (De Gagne, 2023; O'Connor & ChatGPT, 2023; Seibert et al., 2021).

Potential of Generative AI in Nursing Education

Generative AI, like ChatGPT, could aid nursing students in writing and structuring assignments. These generative AI models may assist students in organizing their thoughts and writing. For example, enabling students to use prompts to generate initial ideas can guide the assignment development process. In using generative AI to support brainstorming and idea generation, there is potential to support shifting students' focus towards higher-level critical thinking tasks and reasoning. By offering this support, students can focus more on refining their nursing expertise (Office of Educational Technology, 2023; Sullivan et al., 2023).

AI can also be implemented in complex nursing simulations and decision-making processes. For example, realistic patient scenarios can be generated and used for simulation exercises, enabling nursing students to hone their clinical decision-making skills in a controlled, low-risk environment (Betts et al., 2020; Komasaawa & Yokohira, 2023). A tutoring system powered by generative AI can provide students with personalized and interactive guidance similar to what a human tutor would provide (Bozkurt et al., 2023). Furthermore, these models could provide immediate feedback, thereby personalizing and enhancing the learning process and potentially “learning” from student interactions toward continuous system refinement.

Beyond the potential to contribute in the classroom, AI could also foster personalized learning and enhance the accessibility to study resources. AI can curate learning materials, tailor them to individual needs, and cater to a spectrum of learning styles—from auditory to kinesthetic and from solitary to social—thus enabling a truly adaptive educational experience. Generative AI models could make high-quality educational content more accessible (Bekeš & Galzina, 2023; Neumann et al., 2023; Su & Yang, 2023). For example, entering a prompt in ChatGPT to write a plain language version of a scientific article or using one of many web

browser plugins that use AI to summarize articles (e.g., Scholarcy <https://www.scholarcy.com/>) can serve as complementary tools to support student learning. Further, grades and feedback can be automated for objective assessments, saving teachers time and providing students with immediate feedback.

Challenges with Generative AI in Nursing Education

Despite the potential benefits of generative AI tools in the current digital transformation era, their adoption in nursing education brings new challenges that must be addressed to achieve effective and ethical integration of generative AI in education. Integrating generative AI into nursing education presents several unique challenges, such as academic integrity. With generative AI technologies, like ChatGPT's text generation capabilities, it can be difficult to distinguish AI-generated from student-written content, potentially undermining the assessment and evaluation processes (Dwivedi et al., 2023). A recent empirical study showed that ChatGPT detecting software often introduces an additional bias toward non-native English-speaking students by penalizing students using constrained linguistic expressions (Liang et al., 2023). As demonstrated in the same study by Liang et al. (2023), it is also possible to use simple prompting strategies to effectively bypass ChatGPT detectors. Hence, at this point, automatically created texts are hard to detect.

Critical thinking should represent a core skill nursing students must develop during their educational programs. With the increasing adoption of generative AI, overreliance on generative AI technologies can potentially decrease the effectiveness of student learning and critical thinking development, which may generate unintended consequences for professional practice, including risks to patient safety, privacy, and confidentiality. For example, students may include confidential patient information as prompt inputs to develop more personalized generative AI outputs without knowing whether the system will retain and incorporate these

data. Indeed, large companies such as Samsung and Apple have experienced challenges related to employees uploading proprietary information in ChatGPT, leading to the restriction or banning of the use of generative AI tools in those companies (Kharpal, 2023; Mok, 2023).

The potential for plagiarism and copyright infringement also arises. Generative AI, like ChatGPT, has been trained on vast amounts of online data, some of which might be drawn from copyrighted sources. Consequently, there is a risk that generated text might unintentionally reproduce elements from these sources, raising concerns about plagiarism and copyright infringement (Ray, 2023). Indeed, ChatGPT developer, OpenAI, is currently facing lawsuits and government inquiries about whether the company violated the privacy and copyrights of countless people when its training data were scraped from the internet (Vynck, 2023).

Ethical implications also exist. Generative AI learns from the data they are trained on and can unwittingly reproduce societal biases (e.g., age, sex, race) present in the data, leading to outputs that could reinforce discrimination, stereotypes, or produce biased outcomes (Borenstein & Howard, 2021; Schaich Borg, 2022). Many developers of generative AI do not provide information about the training data they use in the most recent iterations of their generative AI, raising concerns about the extent to which biases exist in the training data and whether attempts to mitigate bias are made (Xiang, 2023). Subsequently, students' uncritical use of these tools without awareness of their potential for producing biased outputs can exacerbate existing health inequities, particularly for populations not well represented in the training data. It is crucial for every user to have the competence to comprehend how AI models work and identify areas where bias risks may arise.

Despite the sophisticated language output of generative AI tools, it is crucial to remember that they do not understand the text they produce. They simply generate text sequences based

on the probability of certain words appearing together. Although this results in text that seems knowledgeable and coherent, the AI does not truly “understand” its output. This could lead to “hallucinations”—instances where the AI produces plausible-sounding but incorrect or unrelated information. Such susceptibility to misinformation could, in turn, lead to erroneous interpretation or application of nursing knowledge, potentially impacting patient care. Consequently, any information generated by AI tools must be rigorously checked for accuracy before being presented as factual in an educational context.

Recommendations for Nurse Educators

Clear guidelines and institutional policies surrounding generative AI use in academic work are critical to mitigate these challenges, and many universities are developing these. Policies should dictate when, how, and by whom generative AI tools can be used and clearly articulate the consequences of their misuse. Comprehensive education on the appropriate use of generative AI by nursing students is essential to enhance their digital literacy skills. Students need to understand how to use these tools effectively and their limitations and potential risks. Students must be aware of issues such as algorithmic bias, the risk of misinformation, and possible copyright infringement or plagiarism (Neumann et al., 2023; O'Connor & Booth, 2022). Nursing schools must recognize that a prerequisite to support nursing students' awareness of generative AI is that nurse educators also have a foundational understanding. Critical thinking and ethical reasoning should be at the core of the nursing curriculum, particularly concerning generative AI. Case studies exploring the ethical implications of generative AI use and active discussions on how these new digital tools can be responsibly used in nursing practice can be incorporated. Lastly, generative AI can be harnessed for administrative tasks, such as grading multiple-choice exams or helping to organize course schedules. This allows educators to focus on core teaching duties,

embracing the potential of AI while addressing its challenges. Table 1 provides concrete examples of ChatGPT's applications in nursing education across three focus areas:

- (1) Idea Generation Promoting Critical Thinking;
- (2) Personalized Learning; and
- (3) Increasing Instructor Efficiency.

Additionally, the table incorporates “Responsible Use Considerations” to offer insights into the limitations and ethical dimensions of using ChatGPT in each scenario.

Table 1. Concrete Examples of Utilizing ChatGPT in Nursing Education and Responsible Use Considerations

Focus Area	Example	Description	Responsible Use Considerations
Idea generation promoting critical thinking	Clinical scenario simulation	Using a technique known as “Persona Prompt,” a nurse educator can instruct ChatGPT to simulate an older adult patient with heart failure. Nursing students will interact with this artificial patient to conduct assessments and develop individualized treatment plans, thereby fostering critical thinking skills	Although simulation can be valuable, instructors should remind students that ChatGPT cannot replace real clinical experiences. Additionally, the AI-generated scenarios may not capture the full complexity of human health conditions

Focus Area	Example	Description	Responsible Use Considerations
Personalized learning	Targeted pharmacology learning	ChatGPT can focus on specific classes of medications, such as antihypertensives, to adapt to the student's individual learning needs. For instance, students can engage in dialogue with ChatGPT to learn about the mechanism of action, dosage, and contraindications of beta-blockers, facilitating a more personalized learning experience	Instructors should be cautious about the accuracy of medical information and must validate any AI-generated content. Tailoring the learning process this way also requires understanding the AI's limitations in adapting to educational objectives
Increasing instructor efficiency	Assisting in literature review	ChatGPT or emerging tools such as Elicit (https://elicit.com/) can be trained to perform preliminary literature searches focusing on randomized controlled trials on specific topics, such as effective interventions for weight loss. The tool can summarize findings and identify gaps in existing research, thereby expediting the literature review process for instructors	Although AI can assist in literature review, it is not a substitute for in-depth scholarly investigation. Instructors and students should rigorously verify AI-generated summaries for academic rigor and relevance

Conclusion

The advent of generative AI in educational settings necessitates strategic planning for optimal integration, particularly in nursing education. The potential of AI to enhance learning outcomes, boost academic integrity, and promote education to all is significant. AI's capabilities to generate well-structured texts, assist in complex decision-making processes, and provide personalized learning experiences could greatly advance nursing education (De Gagne, 2023). However, generative AI's effective and responsible use depends on our ability to navigate the accompanying challenges. Addressing concerns around academic integrity, potential plagiarism, copyright infringements, and ethical implications is paramount. A strategic approach that includes clear institutional policies, comprehensive student education, and AI content detection tools could benefit nursing students and educators. Nurse educators are thus called upon to embrace the promise of generative AI and to lead the charge in responsibly harnessing these powerful digital tools. The potential rewards of this endeavour—enhanced learning and increased efficiency—are well worth the challenges we must navigate to realize them.

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